

ILHA FERNANDO DE NORONHA, Brazil

WMO: 824000

Lat: 3.855S

Lon: 32.423W

Elev: 193

StdP: 14.59

Time Zone: -2.00 (W02)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	73.2	74.0	67.7	102.6	77.8	68.2	104.3	77.9	18.7	79.0	17.3	78.7	7.0	150	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.3	88.1	78.4	87.7	78.4	86.9	78.1	80.1	86.0	79.4	85.4	79.1	85.2	9.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	148.7	84.6	77.4	143.6	83.5	77.2	142.7	83.3	43.8	85.8	43.1	86.2	42.8	85.5	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 16.5	14.5	13.8	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	65.3	83.9	2.6	2.1	63.4	85.4	61.8	86.6	60.4	87.7	58.4	89.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.9	81.9	81.7	81.6	80.8	81.5	80.5	79.6	79.3	80.0	80.7	81.4	82.0
	DBStd	1.55	1.31	1.54	1.56	1.79	1.21	1.33	1.24	1.14	0.92	1.01	1.08	0.85
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11287	990	889	980	925	975	915	916	909	900	953	943	992
	CDD65	5812	525	469	515	475	510	465	451	444	450	488	493	527
	CDH74	53679	5256	4739	5137	4419	5012	4225	3650	3449	3705	4296	4625	5166
	CDH80	12944	1440	1277	1372	1044	1245	882	650	588	787	1047	1220	1392
(14) Wind	WSAvg	8.9	8.7	7.9	7.3	6.6	7.6	8.6	9.1	10.1	10.3	10.4	10.4	10.0
(15) Precipitation	PrecAvg	39.3	2.4	4.1	6.7	9.8	6.9	3.9	2.5	0.9	0.4	0.2	0.3	1.2
	PrecMax	59.3	7.6	11.3	13.0	18.5	17.0	15.9	7.2	2.4	1.1	0.7	1.5	6.1
	PrecMin	15.5	0.2	0.4	1.7	4.8	0.9	0.5	0.3	0.3	0.1	0.1	0.0	0.1
	PrecStd	11.8	2.0	2.4	3.0	3.7	3.9	3.2	1.6	0.6	0.2	0.2	0.4	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	89.2	88.2	88.2	87.9	86.4	86.0	85.3	86.4	87.4	88.0	88.1
		MCWB	78.1	78.1	78.6	79.4	79.4	78.4	76.5	75.6	75.9	76.8	77.1	77.8
	2%	DB	87.8	87.9	87.8	87.1	86.4	86.0	84.6	84.5	85.8	86.3	86.7	87.5
		MCWB	78.1	78.3	78.7	79.2	78.8	78.2	76.1	75.5	75.8	76.5	77.0	78.1
	5%	DB	86.9	86.8	86.7	86.2	86.2	85.0	84.3	84.1	84.5	85.8	86.2	86.4
		MCWB	77.8	78.2	78.6	79.0	78.7	77.5	76.0	75.5	75.5	76.3	76.7	77.4
	10%	DB	86.1	86.1	86.1	85.6	85.7	84.3	83.6	82.9	84.1	84.6	85.7	86.0
		MCWB	77.5	78.0	78.6	78.8	78.4	77.0	75.8	75.2	75.4	75.6	76.5	77.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.4	80.1	80.7	80.8	80.6	79.2	78.3	77.7	78.1	79.0	78.8	79.5
		MCDB	86.6	86.6	86.6	86.5	86.3	85.0	82.8	83.3	84.4	85.3	86.8	86.5
	2%	WB	78.9	79.1	79.8	79.9	79.6	78.6	77.4	76.9	76.9	77.5	77.8	78.7
		MCDB	86.2	85.8	85.8	85.4	85.4	84.5	82.7	82.9	83.9	84.8	85.8	85.9
	5%	WB	78.2	78.7	79.1	79.2	79.1	78.1	76.9	76.3	76.3	76.8	77.2	78.1
		MCDB	85.3	85.4	85.2	84.7	84.9	83.9	82.6	82.3	83.3	84.3	85.2	85.0
	10%	WB	77.7	78.2	78.7	78.8	78.6	77.6	76.4	75.8	75.7	76.1	76.7	77.4
		MCDB	84.5	84.4	84.7	84.4	84.3	83.2	81.7	81.8	82.7	83.3	84.3	84.1
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.3	9.2	9.0	8.9	8.5	8.5	8.5	9.2	9.6	9.7	9.2
		MCDBR	9.9	10.1	10.1	10.0	9.4	9.2	9.0	9.0	9.5	10.1	9.9	9.4
	5% WB	MCWBR	4.3	4.6	4.8	5.4	4.8	4.7	4.5	4.6	4.6	4.7	4.3	4.3
		MCDWR	9.4	9.3	9.3	9.5	9.0	8.9	8.5	8.5	9.1	9.6	9.9	9.1
(40) Clear-Sky Solar Irradiance	taub	taub	0.410	0.417	0.412	0.401	0.401	0.398	0.400	0.413	0.437	0.408	0.400	0.414
		taud	2.508	2.506	2.537	2.596	2.576	2.565	2.551	2.481	2.377	2.472	2.516	2.478
	Ebn at Noon	Ebn at Noon	294	292	289	285	277	273	275	278	278	291	295	291
		Edh at Noon	36	36	35	31	31	30	31	35	40	37	35	36
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2020	1964	1920	1728	1729	1706	1760	1964	2101	2194	2201	2099
	RadStd	RadStd	141	148	135	127	109	103	100	50	59	39	38	83

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page